

Philosophy of Space and Time

Lecture 7: *Eliminating Unobservable Structure*

Chris Smeenk

Department of Philosophy, UCLA · April 21, 2026

Today's Plan

- 1 Recap: The State of the Debate
- 2 Absolute Velocity and Galilean Relativity
- 3 Geroch on Spacetime
- 4 Spacetime Diagrams
- 5 Full Newtonian Spacetime
- 6 Symmetry Groups
- 7 Toward Galilean Spacetime
- 8 Summary and Preview

Where We Left Off

Last week we arrived at a diagnosis of Newton's position:

- The **bucket** and **globes** show that absolute *acceleration* has detectable effects
 - Leibniz's **shift arguments** show that Newton postulates more structure than his physics needs: absolute position and absolute velocity are unobservable
- ??** *Can we find the “Goldilocks” amount of structure—enough to support inertia and acceleration, but without the extras?*

Today's Question

Maudlin, Chapter 3

“Is there any way to somehow purge these unobservable absolute velocities from Newton's theory yet keep all of its explanatory power?”

Yes — but it requires rethinking our picture of space and time, using the idea of **spacetime**.

Two stages today:

1. Clarify *exactly* what is unobservable and why (Maudlin, Ch. 3, first half)
2. Introduce **spacetime diagrams** as a tool for seeing the structure

Absolute Velocity: The Real Problem

Position vs. Velocity: Two Different Problems

Leibniz's *static shift* (relocate everything in space) and *kinematic shift* (set everything in uniform motion) look similar, but Maudlin argues they are quite different:

Static Shift

- Move everything 3 miles east
- Maudlin's reply: the actual and counterfactual situations differ, but clear which is actual
- Not a problematic case of *unobservable*

Position vs. Velocity: Two Different Problems

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Kinematic Shift

- Set everything in uniform motion
- Right now, you have a definite absolute velocity (according to Newton)
- But no experiment can determine what it is
- Genuinely **unobservable** facts

Why Absolute Velocity Is Special

Maudlin's key point:

- According to Newton, right now you have a definite absolute velocity—a specific speed in a specific direction through absolute space
 - You might be at absolute rest, or moving at a million miles per hour toward Alpha Centauri
 - By Corollary V (Galilean relativity), no experiment can tell you which
 - This is not a limitation of our instruments—it follow from the *laws*
- ?? *Newton's theory, together with his Laws of Motion, commits him to the existence of experimentally undeterminable physical facts. Should this give us pause?*

Galilean Relativity Is Not Full Relationism

- Galilean relativity: uniform *velocity* is undetectable, but it does *not* imply that “all motion is relative”
- **Acceleration** is still absolute (as Newton establishes in the Scholium)

What We Want

An account of the structures of space and time such that acceleration is absolute but velocity is not.

Can We Just Delete Absolute Velocity?

Newton's framework: absolute space persists through time → bodies have definite trajectories through absolute space → bodies have definite velocities.

- Not obvious how to “erase” absolute velocity from Newton's picture
- If points of space persist through time, then there *is* a fact about whether a body stays at the same point—and hence a fact about its velocity
- To eliminate absolute velocity, we need to give up the persistence of spatial points through time

?? *But if space doesn't persist through time, how can we even talk about motion? This is where the concept of spacetime is helpful.*

Spacetime: Geroch's Framework

From Space and Time to Spacetime

So far we have been talking about *space* and *time* as separate entities. Geroch (Chs. 1–2) introduces a different starting point:

The Basic Concept

Spacetime is the collection of all *events*—all the things that happen at a particular place and time.

- An **event** is maximally specific: “this flash of light, here, now”
- Events are the raw data of physics; everything else defined based on structure we impose on them
- Both “where” and “when” are folded into a single four-dimensional entity

This shift in perspective turns out to be enormously powerful.

Why Start with Events?

Starting with events is neutral between Newton and his critics:

- Newton's picture: spacetime = absolute space $\times$ absolute time, with a fixed way of slicing spacetime into “space at an instant”
- A relationist might accept the collection of events but deny that they are embedded in an independently existing container

(Note: Einstein will later show that spacetime cannot be neatly separated into space and time – so we really need spacetime)

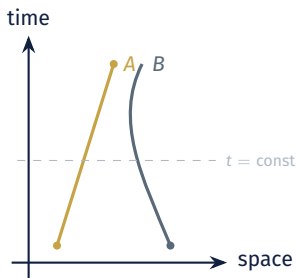
?? *By starting with events and asking “what structure must we add?” Ger-
roch lets us compare different theories of space and time on equal
footing.*

Observers and Worldlines

An **observer** traces out a continuous path through spacetime—a **worldline**.

- Each point on the worldline is an event in the observer's history
- The observer carries a clock that records the elapsed time along their worldline
- Different observers may disagree about which events are “simultaneous”

In Newtonian physics, there is a *fact* about simultaneity (absolute time). In relativistic physics, there is not.



What Structure Does Spacetime Need?

Geroch's framework lets us pose the central question precisely:

- Start with the bare collection of events (a set with topology)
- Add structure as needed:
 - › A notion of **simultaneity** (which events happen “at the same time”)?
 - › A **temporal metric** (how much time elapses between events)?
 - › A **spatial metric** (how far apart are simultaneous events)?
 - › A notion of **straight worldline** (which observers are inertial)?

Newton's spacetime has *all* of these, plus more (absolute rest, absolute velocity).
The question raised by the Leibniz-Clarke debate: can we get by with less?

Newtonian Spacetime in Geroch's Terms

In Geroch's framework, Newton's spacetime has the structure:

Absolute **simultaneity**: a fixed slicing into “instants”

Temporal **metric**: how much time elapses between instants

Spatial **metric**: Euclidean distance within each instant

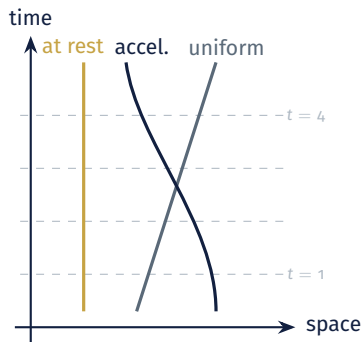
Persistence of spatial points through time (“same place”)

Absolute **velocity**: a fact about which worldlines are “at rest”

Spacetime Diagrams

Building a Spacetime Diagram

1. Take snapshots of space at successive instants
2. Stack them vertically, with later times higher
3. Each body traces a continuous path—a **worldline**
4. A body at rest has a vertical worldline
5. A body in uniform motion has a straight, tilted worldline
6. An accelerating body has a curved worldline



What Spacetime Diagrams Show

- The *angle* of a worldline represents the body's velocity: vertical = rest, tilted = moving
 - A *straight* worldline = inertial (uniform) motion
 - A *curved* worldline = accelerated motion
 - Two worldlines can be *parallel* (same velocity) or converging/diverging
- ?? *Key insight: in the spacetime picture, the distinction between rest and uniform motion is a distinction between directions—between which straight lines count as “vertical.” If we remove the preferred vertical, we remove absolute velocity.*

Full Newtonian Spacetime

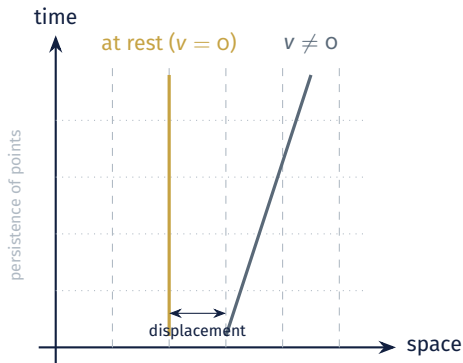
Newton's Spacetime: The Full Picture

In Newton's framework, spacetime has all of the following structures:

1. **Simultaneity**: a fixed slicing into instants (absolute time)
2. **Temporal metric**: a definite elapsed time between any two instants
3. **Spatial metric**: Euclidean distance within each instant
4. **Affine structure**: a fact about which worldlines are straight (inertial)
5. **Persistence of space**: a fact about which events at different times are “at the same place”—equivalently, a preferred notion of **absolute rest**

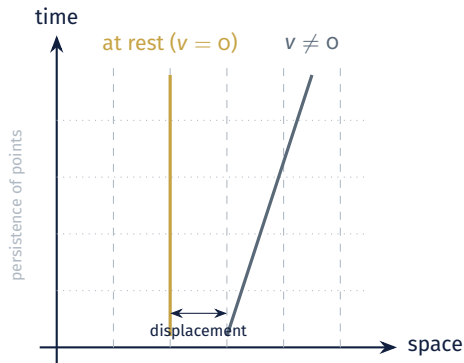
Structure (5) is what gives us absolute velocity. It picks out a preferred “vertical” direction in the spacetime diagram.

Full Newtonian Spacetime: Diagram



The dashed vertical lines represent the persistence of spatial points through time. The gold worldline (vertical) is at absolute rest; the gray one has nonzero absolute velocity.

Full Newtonian Spacetime: Diagram



The dashed vertical lines represent the persistence of spatial points through time. The gold worldline (vertical) is at absolute rest; the gray one has nonzero absolute velocity. Removing the vertical, then no ability to distinguish rest from uniform motion; but straight vs. curved distinction intact.

Symmetries of Full Newtonian Spacetime

The symmetry transformations of full Newtonian spacetime are:

- Spatial translations (shift everything in space)
- Spatial rotations (rotate all of space)
- Time translations (shift the origin of time)

Notably *absent*: velocity boosts. In full Newtonian spacetime, boosting everything to a different velocity gives a *different* state of affairs.

The Mismatch

?? *But Corollary V tells us the physics is the same after a boost. So the physics has more symmetry than the spacetime Newton postulates. This mismatch is exactly the problem.*

A Methodological Principle

The symmetries of a physical theory's spacetime should match the symmetries of its dynamics. If the dynamics have a symmetry that the spacetime lacks, the spacetime postulates too much structure.

- Newton's dynamics are invariant under velocity boosts (Corollary V)
- Full Newtonian spacetime is *not* invariant under boosts
- The dynamics can't "see" which worldlines are at absolute rest
- Therefore: the persistence of spatial points—and hence absolute velocity—is excess structure

Interlude: Symmetry

What Is a Symmetry?

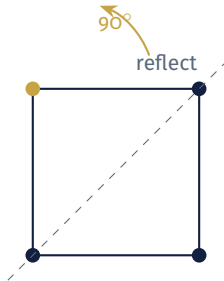
Start with a familiar example: a square.

A **symmetry** of an object is a transformation that leaves it looking exactly the same.

- Rotate a square by $90^\circ \rightarrow$ looks the same
- Rotate by $45^\circ \rightarrow$ looks different
- Reflect across a diagonal $\rightarrow$ looks the same

Two ingredients every time:

1. **Transformation** (rotation, reflection, translation...)
2. Transformation **leaves some features unchanged**



The colored corner tracks what happens: after 90° rotation, a different corner is in its place—but the square as a whole is indistinguishable.

Symmetries of Space

A symmetry of space is a transformation that preserves its *geometric structure*—all distances and angles remain the same.

- **Translation**: slide everything 3 meters to the right. All distances between points are unchanged. Space is *homogeneous*—no point is geometrically special.
- **Rotation**: turn everything 30° around an axis. All distances and angles preserved. Space is *isotropic*—no direction is geometrically special.
- **Reflection**: mirror everything across a plane. Distances preserved again.

Each of these is an **isometry**: a mapping of space onto itself that preserves all distances.

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Compare with Aristotle's spherical universe: it is *not* homogeneous (the center is special), so translations are *not* symmetries. Less symmetry = more structure.

The Symmetry Group

The symmetries of an object (or a space) form a **group**—a collection of transformations with a simple algebraic structure:

1. **Closure**: if A and B are symmetries, then doing A followed by B is also a symmetry
2. **Identity**: doing nothing at all is always a symmetry
3. **Inverse**: every symmetry can be undone (e.g., rotating by 90° is reversed by rotating by -90°)

The Key Idea

The symmetry group of a space *encodes* its geometric structure. **More symmetries = less structure; fewer symmetries = more structure.**

Symmetry Groups and Spacetime

This idea will be our main tool for comparing spacetimes:

- Each spacetime has a symmetry group: the set of all transformations that preserve its structure
- More structure $\rightarrow$ fewer symmetries (a smaller group)
- Less structure $\rightarrow$ more symmetries (a larger group)

We will see:

- **Full Newtonian** spacetime: translations + rotations (no boosts)—*smaller* group
- **Galilean** spacetime: translations + rotations + *velocity boosts*—*larger* group

?? *Galilean spacetime has a larger symmetry group because it has less structure—it lacks the “preferred rest frame” that full Newtonian spacetime has.*

Break

Toward Galilean Spacetime

What to Keep, What to Discard

Simultaneity: which events happen “at the same time” ✓

Temporal metric: elapsed time between instants ✓

Spatial metric: Euclidean distance on each time slice ✓

Affine structure: which worldlines are straight (inertial) ✓

Persistence of spatial points / absolute velocity ✗

Discard the bottom layer → **Galilean spacetime** (also called *neo-Newtonian spacetime*).

The Key Idea

In Galilean spacetime:

- There is *no fact* about whether two events at different times occur “at the same place,” or about whether a body is at absolute rest or in uniform motion
- But there *is* a fact about which worldlines are straight (inertial) and which are curved (accelerating)

?? *This sounds paradoxical: how can there be absolute acceleration without absolute velocity? After all, acceleration is the rate of change of velocity. If there is no absolute velocity, what is changing?*

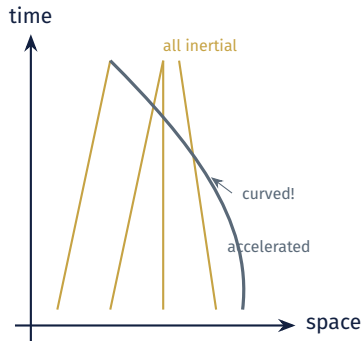
The answer: think geometrically. Acceleration = **bending of a worldline** in spacetime. Bending is an intrinsic geometric property—it doesn't require a preferred “vertical.”

Acceleration as Curvature of Worldlines

In spacetime, the distinction between inertial and accelerated motion is *geometric*:

- A straight worldline = no acceleration = no force
- A curved worldline = acceleration = force acting
- You can tell whether a worldline is straight or curved *without* choosing a preferred direction for “at rest”

(Like asking whether a road on the earth’s surface is straight or curved—you don’t need a compass direction to answer.)



Questions That Make Sense in Galilean Spacetime

Meaningful questions:

- How much time elapsed between two events?
- Are two events simultaneous?
- How far apart are two simultaneous events?
- Is a worldline straight or curved?
- What is the relative velocity of two bodies?

The “meaningless” questions are exactly the ones whose answers Newton’s dynamics cannot determine.

Questions without answers:

- Are two non-simultaneous events at the same place?
- What is a body’s absolute velocity?
- Is a body at absolute rest?
- How far apart are two events that occur at different times?

Summary

1. **Absolute velocity** is genuinely unobservable: Newton's own Corollary V guarantees this. This is a deeper problem than the static shift.
2. In **full Newtonian spacetime**, a preferred vertical direction (persistence of spatial points) gives absolute velocity; this structure has no dynamical counterpart.
3. The **mismatch principle**: spacetime symmetries should match dynamical symmetries. Newton's spacetime fails in this respect.
4. **Galilean spacetime** eliminates the persistence of spatial points, retaining everything else.
5. Acceleration = **curvature of a worldline**—an intrinsic geometric property that does not require absolute velocity.

For Thursday

- **Read:** Maudlin, Ch. 3 (finish); Geroch, Chs. 3–4.
- **Think about:**
 - › What does it mean to say that two spacetime diagrams “represent the same physical situation” in Galilean spacetime?
 - › How does Geroch’s treatment of observers and particles relate to the spacetime picture?

Thursday: **Galilean Spacetime in Detail**

- The Galilean boost and what it preserves
- Newton’s arguments revisited in the spacetime picture
- Geroch on particles, observers, and the structure of spacetime
- Looking ahead: what happens when we give up simultaneity too?

Questions?

chris.smeenk@gmail.com